

Week 9: Bayesian regression (Lab Answers)

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1 Packages required

```
library(rstan)
library(tidyverse)
library(tidybayes)
library(janitor)
```

2 Reminder: what is a Gompertz model?

In 1825 Benjamin Gompertz, an actuary in London, proposed the following two-parameter model for mortality:

$$\mu(x) = \alpha e^{\beta x}$$

where $\mu(x)$ is the instantaneous mortality rate (hazard) at age x . This model assumes that mortality increases exponentially with age, which is a pretty good assumption for adult mortality (above age 40 or so, with the exception of older ages).

3 Data

For the following two examples we're going to use death and population counts by age and sex for Ontario, sourced from the Canadian Human Mortality Database project. We can read the data files in directly from the URLs:

```
dd <- read_table("https://www.prdh.umontreal.ca/BDLC/data/ont/Deaths_1x1.txt", skip = 1)
dp <- read_table("https://www.prdh.umontreal.ca/BDLC/data/ont/Population.txt", skip = 1)
```

These data files are in 'wide' format. For our purposes it's going to be easier to work with in 'long format'. So let's do that and also clean some other stuff up:

```

dd <- dd |>
  clean_names() |>
  pivot_longer(-(year:age), names_to = "sex", values_to = "deaths") |>
  mutate(deaths = as.numeric(deaths), age = as.numeric(age)) |>
  mutate(age = ifelse(is.na(age), 110, age))

dp <- dp |>
  clean_names() |>
  pivot_longer(-(year:age), names_to = "sex", values_to = "pop") |>
  mutate(pop = as.numeric(pop), age = as.numeric(age)) |>
  mutate(age = ifelse(is.na(age), 110, age))

d <- dd |>
  left_join(dp) |>
  mutate(mx = deaths/pop,
         log_mx = log(mx))

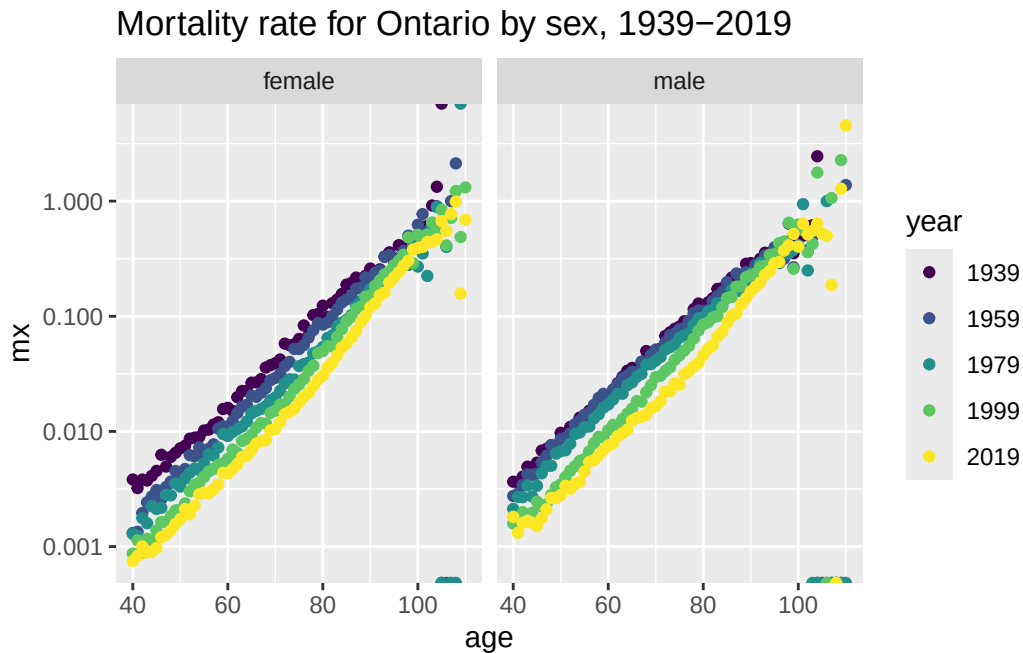
```

Do some quick plots

```

d |>
  filter(age>39, year %in% seq(1939, 2019, by = 20), sex!= "total") |>
  ggplot(aes(age, mx, color = factor(year))) +
  geom_point() +
  facet_wrap(~sex)+
  labs(title = "Mortality rate for Ontario by sex, 1939-2019")+
  scale_y_log10()+
  scale_color_viridis_d(name = "year")

```



4 Base model

First let's fit a Gompertz model to males aged 40+ in 2019. Note that Gompertz models have the form

$$\mu(x) = \alpha e^{\beta x}$$

So we can write

$$\log \mu(x) = \log \alpha + \beta x$$

More notes:

- To make this fully Bayesian we need to specify the likelihood and priors. The full model here is

$$y_x \sim \text{Poisson}(P_x \cdot m_x)$$

$$\log m_x = \log \alpha + \beta x$$

$$\log \alpha \sim N(0, 10^2)$$

$$\beta \sim N(0, 0.1^2)$$

where y_x is deaths at age x and P_x is population

- Could have used a normal likelihood (c.f. using `lm`) but nice to account for population size
- We are fitting not on age, but on a centered version (why?)

Now we need to get the data in the right format to read into Stan (this required a named list):

```
d_male_19 <- d |>
  filter(year==2019, sex == "male", age>39, age<105) |>
  mutate(age_c = age - 40)

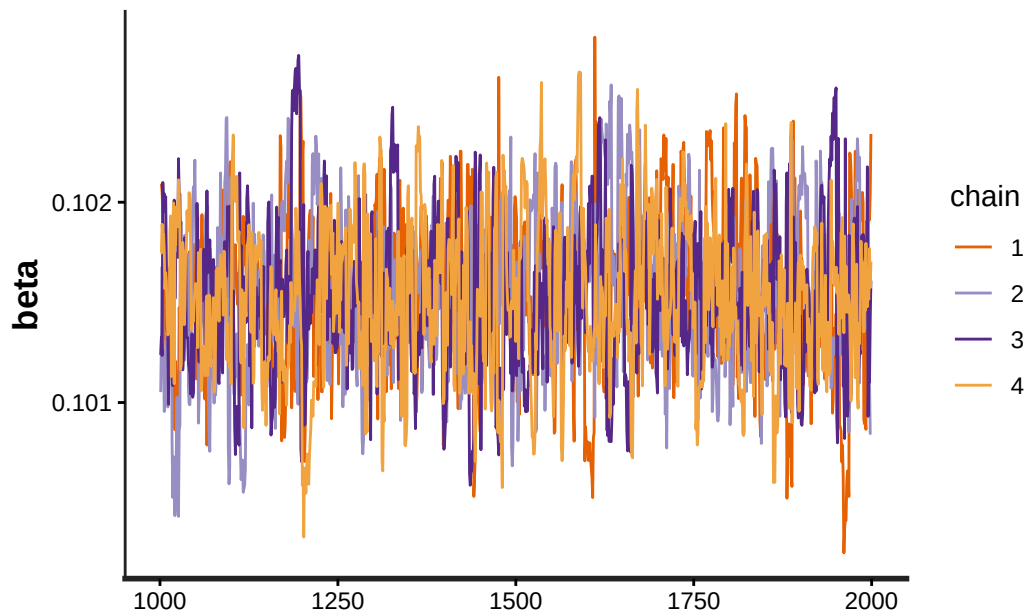
stan_data <- list(y = round(d_male_19$deaths),
                 pop = round(d_male_19$pop),
                 N = nrow(d_male_19),
                 age_c = d_male_19$age_c)
```

Run the model and look at some output:

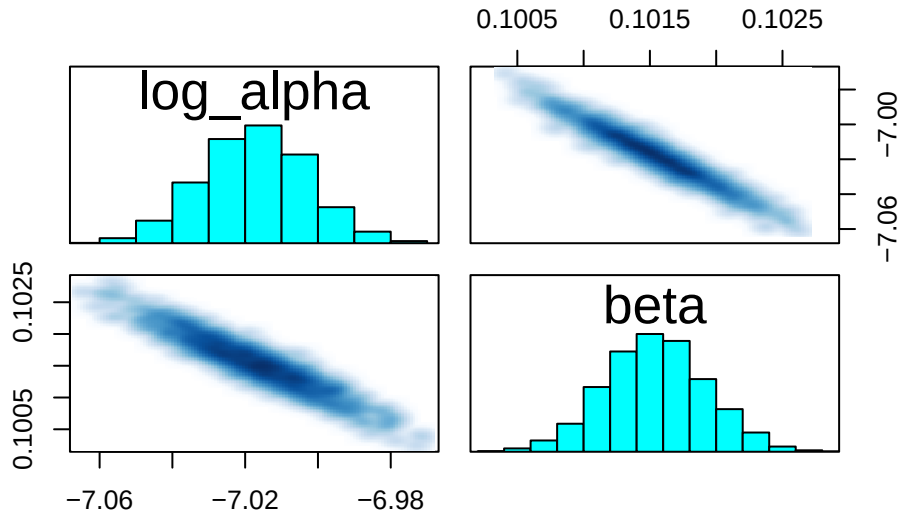
```
mod <- stan(file = "/Users/liangqishen/Desktop/data/gomp.stan",
            data = stan_data,
            seed = 123)
mod
```

Some quick model checks, looking at the traceplot and pairwise densities

```
pars <- c("log_alpha", "beta")
traceplot(mod, pars = c("beta"))
```



```
pairs(mod, pars = pars)
```



```
temp <- rstan::extract(mod)
names(temp)
```

```
[1] "log_alpha" "beta"      "mu"        "lp_"
```

```
temp$log_alpha
```

```
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 [3606] -7.036196 -7.010362 -7.003723 -7.038345 -7.022790 -6.992747 -7.013204
 [3613] -7.008441 -7.032755 -7.002504 -7.034229 -7.028108 -7.034015 -7.049283
 [3620] -7.014801 -7.015153 -7.033732 -7.016423 -7.037513 -7.029396 -7.020208
 [3627] -7.030538 -7.033267 -7.012975 -7.009192 -6.996556 -7.012454 -7.042405
 [3634] -7.011000 -7.027490 -6.994210 -7.041497 -6.979199 -7.019872 -7.009424
 [3641] -7.002118 -6.991651 -7.035537 -7.010085 -7.023542 -6.997337 -7.025759
 [3648] -7.013738 -7.037021 -6.992416 -7.016531 -7.041673 -7.013514 -7.016854
 [3655] -7.021809 -7.004684 -7.033128 -7.031757 -6.997605 -7.020254 -7.042945
 [3662] -7.055434 -7.020721 -7.031831 -6.995190 -7.031900 -7.001426 -7.014363
 [3669] -7.030436 -6.993276 -7.003101 -7.026768 -7.021145 -7.025175 -7.039295
 [3676] -6.987993 -6.999639 -7.008069 -7.026216 -6.987913 -7.004510 -7.028898
 [3683] -6.993439 -7.019281 -7.013614 -6.986403 -7.031393 -7.021661 -7.007884
 [3690] -6.994544 -7.019460 -7.020329 -7.001314 -7.005974 -7.010125 -7.024413

[3697] -7.007406 -7.025412 -6.999619 -7.040105 -7.011650 -7.019867 -7.021605
 [3704] -7.007140 -7.006318 -7.020356 -7.002174 -7.035141 -6.998209 -7.028413
 [3711] -7.018482 -7.059136 -7.037238 -7.034778 -7.019733 -7.009612 -7.024149
 [3718] -7.025339 -7.018798 -7.012964 -7.014688 -7.022399 -7.016215 -7.008294
 [3725] -7.029051 -7.009244 -7.030436 -7.012038 -7.019580 -7.038124 -7.020839
 [3732] -7.000745 -7.028030 -7.041460 -7.031382 -7.056970 -7.009000 -7.002586
 [3739] -7.006076 -7.040512 -7.033176 -6.997641 -7.032838 -7.005733 -7.019467
 [3746] -7.016585 -6.998045 -7.035137 -7.003836 -6.986410 -7.012210 -6.987913
 [3753] -7.039503 -7.032167 -7.035821 -7.000113 -7.015561 -7.020724 -7.019481
 [3760] -7.019575 -7.022262 -7.045744 -7.012933 -7.003658 -7.026580 -7.022499
 [3767] -7.028461 -7.043905 -7.019623 -7.004211 -7.011386 -7.006632 -7.015012
 [3774] -7.033494 -7.026477 -6.988644 -7.017122 -7.012509 -7.002514 -7.015147
 [3781] -7.001360 -7.021917 -7.034907 -7.013594 -7.010947 -7.007140 -7.013688
 [3788] -7.026592 -7.007613 -6.991546 -7.012685 -6.992766 -7.030782 -7.036980
 [3795] -7.028958 -6.984431 -7.029550 -7.016679 -6.995554 -7.039170 -7.005299
 [3802] -7.016040 -7.030592 -7.030286 -7.018134 -7.028773 -7.005862 -7.023624
 [3809] -7.030987 -7.016019 -7.020562 -7.014605 -7.052654 -7.013796 -7.045311
 [3816] -7.009916 -7.032911 -7.014423 -7.017953 -7.016898 -7.027089 -7.020026
 [3823] -7.020997 -7.030632 -7.001121 -7.029881 -7.021001 -7.002375 -7.040414
 [3830] -7.015969 -7.056869 -7.022197 -7.016042 -7.008537 -7.030726 -7.020151
 [3837] -7.031284 -7.024012 -7.019877 -7.021425 -7.022881 -7.025044 -6.990961
 [3844] -7.020473 -7.042867 -7.029097 -7.045028 -7.011545 -7.021305 -6.978446
 [3851] -7.020761 -7.032422 -7.039355 -7.030444 -7.025500 -6.997412 -7.030762
 [3858] -6.998178 -7.038319 -7.000434 -7.039427 -7.010268 -7.023234 -7.009318
 [3865] -7.016665 -7.015293 -7.013719 -7.021481 -7.015609 -6.989496 -7.010523
 [3872] -6.999393 -7.009015 -7.039918 -7.030744 -7.006326 -7.019531 -6.978732
 [3879] -6.990407 -7.013447 -7.004195 -7.006107 -7.014157 -7.022030 -7.004491
 [3886] -7.007576 -6.993579 -7.005974 -7.032473 -7.032629 -6.988858 -7.037172
 [3893] -7.019712 -7.005422 -7.013022 -7.008642 -7.047065 -7.025647 -7.011858
 [3900] -7.003107 -7.025540 -7.026681 -7.000784 -7.008375 -7.031090 -6.992440
 [3907] -7.015827 -7.016883 -7.032420 -7.026715 -7.008676 -7.012179 -7.012242
 [3914] -7.013345 -7.006184 -7.035100 -7.029027 -7.020990 -7.041707 -7.037755
 [3921] -7.026372 -7.025965 -7.034471 -7.019149 -7.010489 -7.035457 -7.013336
 [3928] -6.997354 -7.034871 -7.008188 -6.993150 -7.000384 -7.030733 -7.028956
 [3935] -6.980339 -7.036707 -6.996781 -7.020192 -7.001032 -7.013435 -7.007635
 [3942] -7.004432 -7.000032 -7.025490 -7.023194 -7.018208 -7.012896 -7.008456
 [3949] -7.018849 -7.035509 -7.006207 -7.032029 -6.991890 -7.014518 -7.011185
 [3956] -7.006639 -7.014235 -7.040114 -7.026195 -7.021563 -7.011105 -7.020706
 [3963] -7.009534 -7.010741 -7.004994 -7.022541 -7.033354 -7.019476 -7.019134
 [3970] -6.995620 -7.012007 -7.001928 -7.000144 -7.024191 -7.055565 -7.035221
 [3977] -7.017598 -7.020771 -7.029065 -7.000331 -7.029232 -7.035655 -6.996166
 [3984] -6.990327 -6.985527 -7.001559 -7.012321 -6.982629 -7.000734 -7.006665
 [3991] -7.028609 -7.003510 -7.016804 -7.020821 -7.010482 -7.016104 -7.005073

```
[3998] -7.026751 -7.007317 -7.003100
```

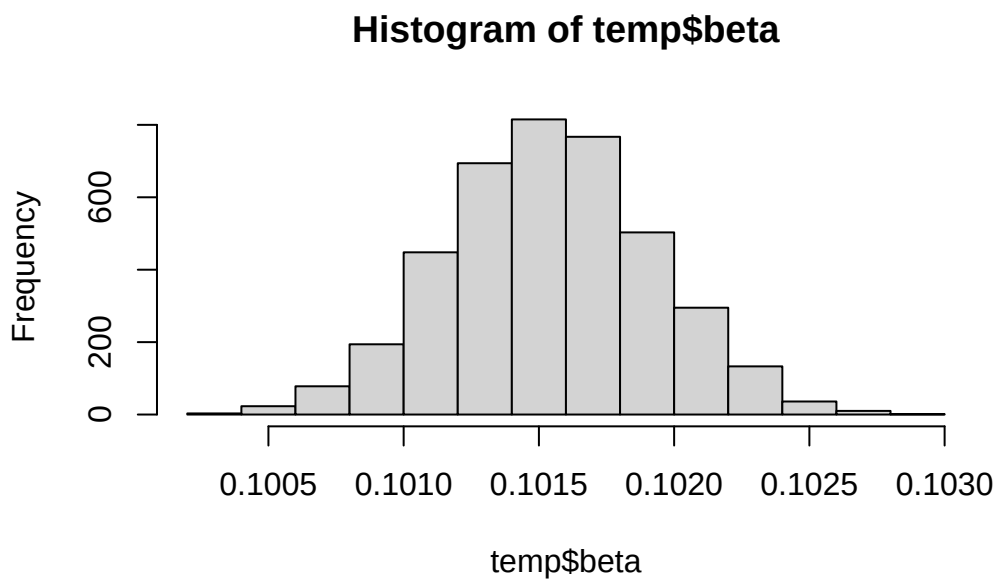
```
median(temp$log_alpha)
```

```
[1] -7.017275
```

```
median(temp$beta)
```

```
[1] 0.1015368
```

```
hist(temp$beta)
```



```
# 95% credible interval for beta  
quantile(temp$beta, 0.025) # lower
```

```
2.5%  
0.1007973
```

```
quantile(temp$beta, 0.975) # upper
```

```
97.5%  
0.1022955
```

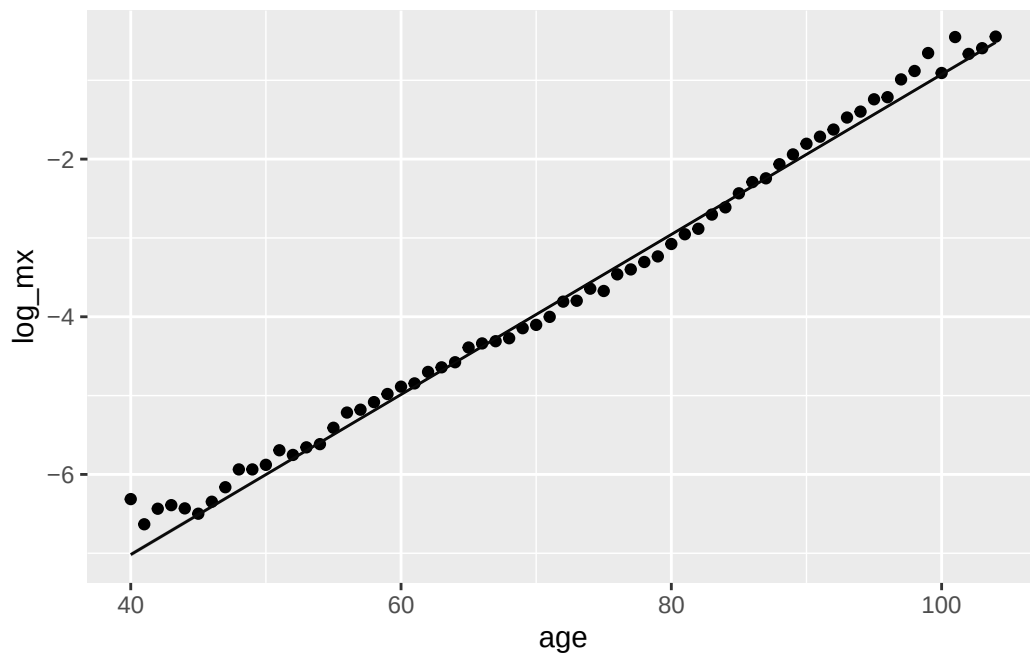
4.1 Extract the parameter estimates using tidyverse

```
mod |>  
  gather_draws(log_alpha, beta) |>  
  median_qi(.width = 0.99)  
  
# A tibble: 2 x 7  
  .variable .value .lower .upper .width .point .interval  
  <chr>      <dbl> <dbl> <dbl> <dbl> <chr> <chr>  
1 beta      0.102  0.101  0.103  0.99 median qi  
2 log_alpha -7.02  -7.06  -6.98  0.99 median qi
```

4.2 Calculate and plot some results

Now we can combine the `tidybayes` syntax with `ggplot` to plot some results. For example, here's the data versus the estimates for the linear predictor:

```
mod |>  
  gather_draws(mu[x]) |>  
  median_qi() |>  
  mutate(age_c = x - 1) |>  
  left_join(d_male_19) |>  
  ggplot(aes(age, log_mx)) +  
  geom_point() +  
  geom_line(aes(age, .value)) +  
  geom_ribbon(aes(x = age, ymin = .lower, ymax = .upper), alpha = 0.2)
```



Let's calculate the modal age at death (which for Gompertz mortality is a function of α and β , see [here](#)).

```
mod |>
  spread_draws(log_alpha, beta) |>
  mutate(mode_age = 1/beta*log(beta/exp(log_alpha))) |>
  median_qi()
```

```
# A tibble: 1 x 12
  log_alpha log_alpha.lower log_alpha.upper beta beta.lower beta.upper mode_age
  <dbl>         <dbl>         <dbl> <dbl>    <dbl>    <dbl>    <dbl>
1    -7.02         -7.05         -6.99 0.102    0.101    0.102    46.6
# i 5 more variables: mode_age.lower <dbl>, mode_age.upper <dbl>, .width <dbl>,
#   .point <chr>, .interval <chr>
```

4.3 Question

Check you get similar results using `lm` or `glm`.

```
summary(lm(log_mx ~ age_c, data = d |>
  filter(!is.infinite(log_mx))) |>
```

```
mutate(age_c = age-40)
))
```

Call:

```
lm(formula = log_mx ~ age_c, data = mutate(filter(d, !is.infinite(log_mx)),
  age_c = age - 40))
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5022	-0.5513	0.0561	0.4002	6.5859

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.6637269	0.0052037	-1088.4	<2e-16 ***
age_c	0.0751362	0.0001534	489.9	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8721 on 33294 degrees of freedom
(608 observations deleted due to missingness)

Multiple R-squared: 0.8782, Adjusted R-squared: 0.8782

F-statistic: 2.4e+05 on 1 and 33294 DF, p-value: < 2.2e-16

```
d |> filter(!is.infinite(log_mx)) |>
  filter(pop>0) |>
  mutate(age_c = age-40) |>
  mutate(deaths = as.integer(deaths),
    pop = as.integer(pop)) |>
  filter(!is.infinite(deaths)) |>
  filter(!is.infinite(pop)) |>
  summary()
```

year		age		sex		deaths	
Min.	:1921	Min.	: 0.00	Length:	33296	Min.	: 0.0
1st Qu.	:1947	1st Qu.	: 26.00	Class	:character	1st Qu.	: 53.0
Median	:1973	Median	: 53.00	Mode	:character	Median	: 155.0
Mean	:1972	Mean	: 53.42			Mean	: 385.4
3rd Qu.	:1998	3rd Qu.	: 80.00			3rd Qu.	: 508.0
Max.	:2023	Max.	:110.00			Max.	:6774.0
pop		mx		log_mx		age_c	

Min. :	0	Min. :	0.0000128	Min. :	-11.270	Min. :	-40.00
1st Qu.:	10190	1st Qu.:	0.0012355	1st Qu.:	-6.696	1st Qu.:	-14.00
Median :	35602	Median :	0.0072344	Median :	-4.929	Median :	13.00
Mean :	49440	Mean :	0.0971173	Mean :	-4.656	Mean :	13.42
3rd Qu.:	75617	3rd Qu.:	0.0851959	3rd Qu.:	-2.463	3rd Qu.:	40.00
Max. :	240864	Max. :	4.5454545	Max. :	1.514	Max. :	70.00

```
summary(glm(deaths ~ age_c + offset(log(pop)),
            family = "poisson", data = d |> filter(!is.infinite(log_mx))
            mutate(age_c = age-40) |>
            mutate(deaths = as.integer(deaths),
            pop = as.integer(pop)) |>
            filter(pop>0) |>
            filter(!is.infinite(deaths)) |>
            filter(!is.infinite(pop))
            ))
```

Call:

```
glm(formula = deaths ~ age_c + offset(log(pop)), family = "poisson",
    data = filter(filter(filter(mutate(mutate(filter(d, !is.infinite(log_mx))),
    age_c = age - 40), deaths = as.integer(deaths), pop = as.integer(pop)),
    pop > 0), !is.infinite(deaths)), !is.infinite(pop)))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.816e+00	5.327e-04	-10917	<2e-16 ***
age_c	7.399e-02	1.611e-05	4593	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 36321057 on 33132 degrees of freedom
 Residual deviance: 7505026 on 33131 degrees of freedom
 AIC: 7730825

Number of Fisher Scoring iterations: 6

5 Model over time

Let's fit a slightly more complicated model, for multiple years, where the coefficients themselves are modeled as a random walk over time, i.e.

$$\beta_t \sim N(\beta_{t-1}, \sigma_\beta^2)$$

That is, a different set of Gompertz parameters are fit to every year, but we are assuming that the values in the current year are related to those in the previous year. This is a form of dynamic linear regression. We need to put priors on the first time point, and also on the variance terms:

$$\log \alpha_1 \sim N(-6, 1)$$

$$\beta_1 \sim N(0.1, 0.1^2)$$

$$\sigma^\alpha, \sigma_\beta \sim N^+(0, 1)$$

Note that the likelihood and model on mortality rates are as before, we just have an additional subscript for time:

$$y_{x,t} \sim \text{Poisson}(P_{x,t} \cdot m_{x,t})$$

$$\log m_{x,t} = \log \alpha_t + \beta_t x$$

Now to fit the model. First get the data in the right format:

```
years <- 1989:2019
d_male <- d |> filter(year>=years[1], sex == "male", age>39, age<105) |>
  mutate(age_c = age - 40)

y <- d_male |>
  select(age, year, deaths) |>
  pivot_wider(names_from = "year", values_from = "deaths") |>
  select(-age) |>
  as.matrix()

pop <- d_male |>
  select(age, year, pop) |>
  pivot_wider(names_from = "year", values_from = "pop") |>
  select(-age) |>
  as.matrix()

stan_data <- list(y = y,
                  pop = pop,
                  N = nrow(d_male_19),
                  age_c = d_male_19$age_c,
                  T = ncol(y))
```


Now fit the model (note: takes a while):

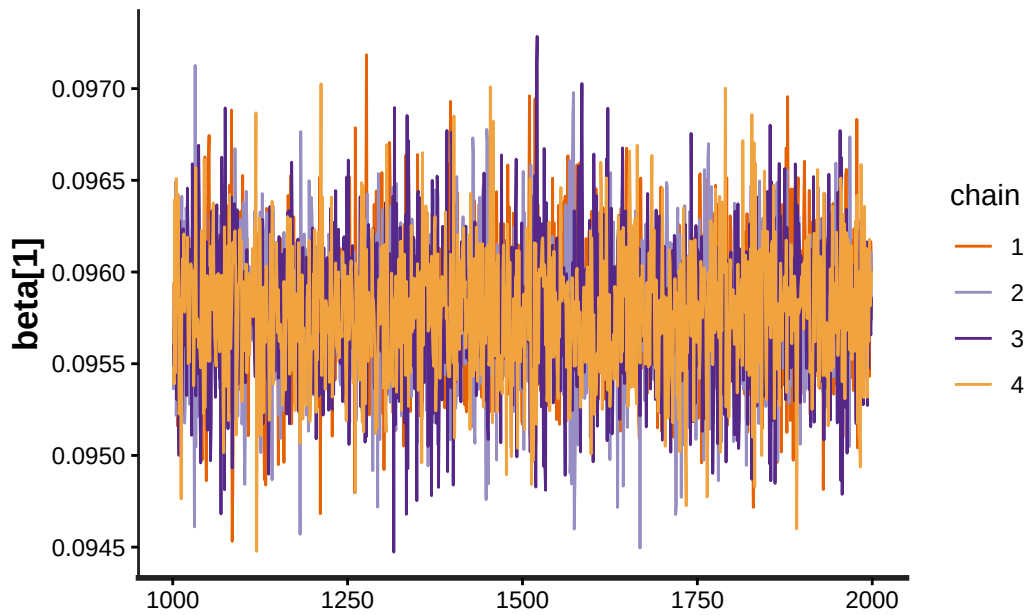
```
mod <- stan(file = "/Users/liangqishen/Desktop/data/gomp_time.stan",
            data = stan_data,
            seed = 852,
            refresh = 0)
```

```
summary(mod)$summary[paste0("beta[", 1:(length(years)), "]",),]
```

	mean	se_mean	sd	2.5%	25%	50%
beta[1]	0.09578071	4.829066e-06	0.0003838707	0.09503805	0.09552147	0.09578169
beta[2]	0.09560984	5.095156e-06	0.0003484006	0.09492545	0.09537648	0.09561422
beta[3]	0.09666680	4.727472e-06	0.0003425171	0.09599732	0.09643687	0.09666527
beta[4]	0.09687345	4.676795e-06	0.0003387142	0.09620980	0.09664365	0.09687475
beta[5]	0.09707949	4.712824e-06	0.0003440414	0.09639913	0.09684559	0.09708075
beta[6]	0.09803747	4.652892e-06	0.0003284409	0.09739200	0.09781290	0.09803919
beta[7]	0.09873206	4.715881e-06	0.0003447994	0.09805308	0.09849860	0.09872808
beta[8]	0.09981707	4.222076e-06	0.0003212879	0.09918531	0.09960228	0.09981500
beta[9]	0.10069572	4.420913e-06	0.0003362704	0.10001407	0.10047159	0.10069793
beta[10]	0.10200652	4.944386e-06	0.0003375353	0.10135490	0.10177519	0.10200577
beta[11]	0.10278784	4.843734e-06	0.0003421987	0.10211910	0.10255204	0.10278587
beta[12]	0.10217584	4.791354e-06	0.0003366845	0.10152847	0.10194695	0.10217943
beta[13]	0.10153793	4.618069e-06	0.0003272754	0.10089573	0.10131247	0.10153769
beta[14]	0.10108387	4.413320e-06	0.0003294230	0.10041604	0.10086530	0.10108472
beta[15]	0.10134993	4.507229e-06	0.0003232743	0.10071363	0.10112957	0.10134601
beta[16]	0.10127177	4.646599e-06	0.0003234779	0.10064646	0.10105332	0.10126850
beta[17]	0.10169917	4.705902e-06	0.0003247575	0.10106519	0.10147804	0.10169699
beta[18]	0.10161182	4.389358e-06	0.0003180507	0.10101262	0.10140347	0.10161350
beta[19]	0.10158293	4.300499e-06	0.0003108208	0.10096464	0.10137545	0.10158716
beta[20]	0.10206954	4.244479e-06	0.0003158327	0.10145299	0.10186080	0.10206731
beta[21]	0.10203856	4.219763e-06	0.0003234555	0.10141347	0.10181900	0.10203876
beta[22]	0.10196812	4.079128e-06	0.0003138058	0.10134820	0.10175087	0.10197044
beta[23]	0.10201856	4.260549e-06	0.0003146841	0.10140961	0.10180429	0.10201179
beta[24]	0.10191453	4.335975e-06	0.0003139596	0.10132160	0.10169663	0.10190919
beta[25]	0.10225546	3.948971e-06	0.0003033437	0.10167323	0.10205376	0.10225769
beta[26]	0.10246939	3.961961e-06	0.0003075654	0.10185663	0.10226882	0.10247081
beta[27]	0.10239959	4.124113e-06	0.0003089095	0.10178123	0.10219091	0.10240284
beta[28]	0.10161254	4.056159e-06	0.0003083612	0.10101623	0.10140396	0.10160557
beta[29]	0.10211345	4.207566e-06	0.0002938204	0.10154192	0.10191555	0.10211034
beta[30]	0.10173138	4.089656e-06	0.0002933123	0.10116200	0.10153923	0.10173113
beta[31]	0.10119351	4.254486e-06	0.0002904419	0.10063940	0.10099141	0.10119457
	75%	97.5%	n_eff	Rhat		

beta[1]	0.09603772	0.09653981	6318.931	1.0000508
beta[2]	0.09584654	0.09629612	4675.657	0.9991337
beta[3]	0.09689639	0.09734534	5249.363	0.9992319
beta[4]	0.09710453	0.09752886	5245.297	0.9993043
beta[5]	0.09731163	0.09772957	5329.162	1.0000540
beta[6]	0.09826012	0.09868084	4982.743	0.9992321
beta[7]	0.09897246	0.09940785	5345.732	0.9998854
beta[8]	0.10003882	0.10043723	5790.777	0.9991640
beta[9]	0.10091820	0.10135112	5785.666	0.9994756
beta[10]	0.10223431	0.10267774	4660.298	0.9994327
beta[11]	0.10301839	0.10345286	4991.098	0.9995251
beta[12]	0.10240900	0.10282772	4937.757	1.0000330
beta[13]	0.10175913	0.10217073	5022.337	0.9994743
beta[14]	0.10130676	0.10171796	5571.562	0.9995331
beta[15]	0.10156724	0.10198806	5144.263	0.9999017
beta[16]	0.10148538	0.10191820	4846.397	0.9996614
beta[17]	0.10191531	0.10233626	4762.475	0.9994138
beta[18]	0.10181937	0.10221700	5250.381	1.0001158
beta[19]	0.10178828	0.10220077	5223.752	0.9996767
beta[20]	0.10228471	0.10268153	5536.882	0.9994306
beta[21]	0.10225582	0.10267611	5875.611	0.9996466
beta[22]	0.10218204	0.10258265	5918.165	0.9994581
beta[23]	0.10223093	0.10263351	5455.298	0.9993488
beta[24]	0.10212719	0.10253073	5242.929	0.9998571
beta[25]	0.10245584	0.10285768	5900.681	0.9993644
beta[26]	0.10267688	0.10306646	6026.354	0.9995832
beta[27]	0.10261110	0.10299813	5610.497	0.9996403
beta[28]	0.10181824	0.10222448	5779.491	0.9999550
beta[29]	0.10231783	0.10269072	4876.431	0.9993700
beta[30]	0.10192729	0.10230177	5143.833	0.9998659
beta[31]	0.10138856	0.10176966	4660.412	0.9999630

```
traceplot(mod, pars = c("beta[1]"))
```

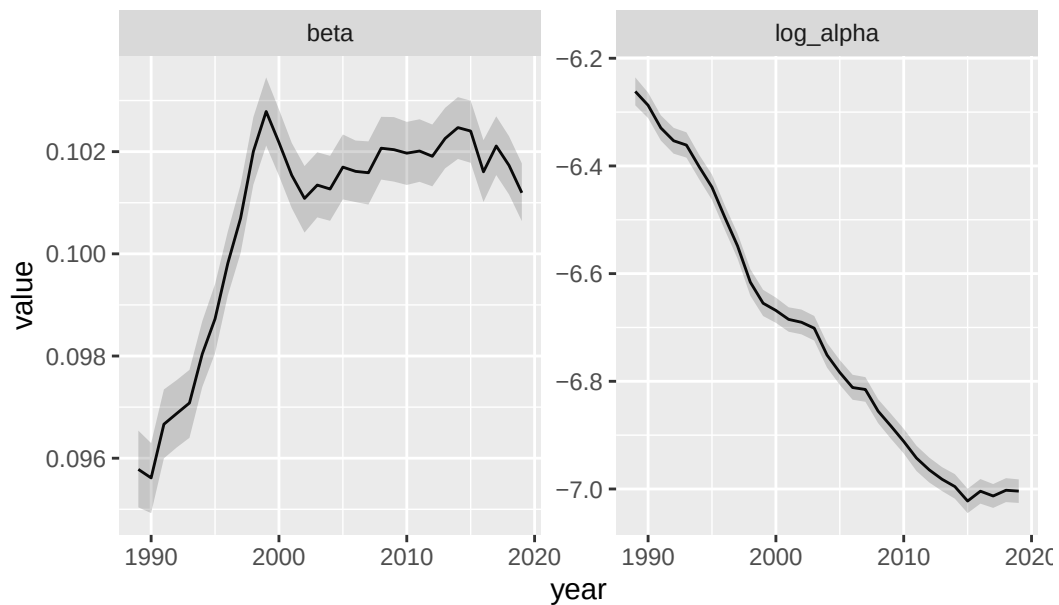


5.1 Plot parameter estimates over time

Now we can plot the parameter estimates over time:

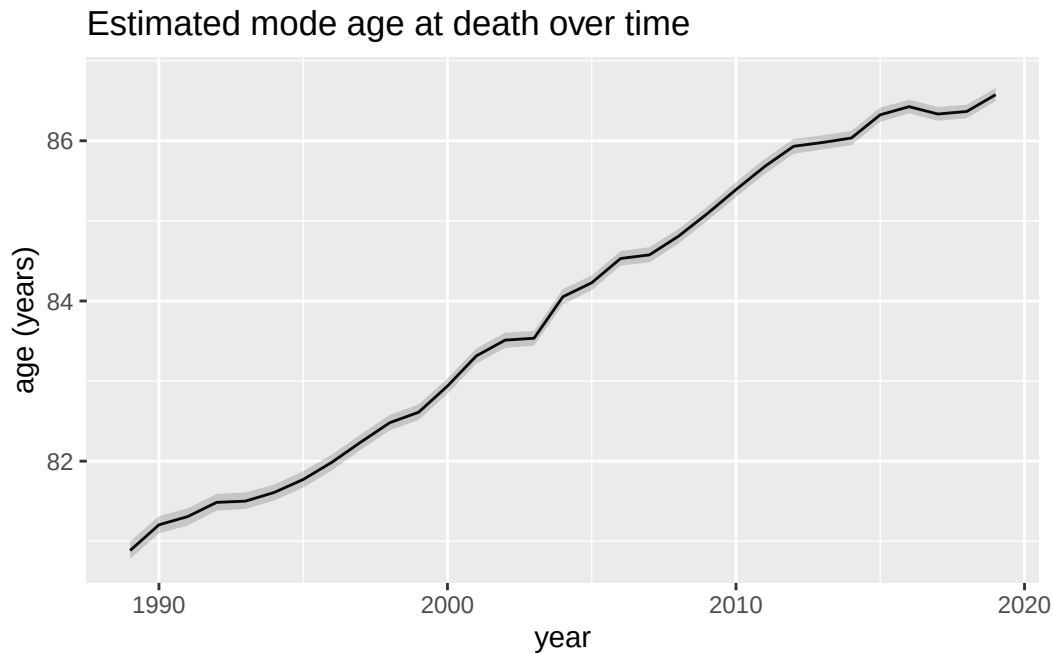
```
mod |>
  gather_draws(log_alpha[i], beta[i]) |>
  median_qi() |>
  mutate(year = years[i]) |>
  ggplot(aes(year, .value)) + geom_line() +
  facet_wrap(~.variable, scales = "free_y") +
  geom_ribbon(aes(ymin = .lower, ymax = .upper), alpha = 0.2)+
  labs(y = "value", title = "Estimates of Gompertz parameters over time")
```

Estimates of Gompertz parameters over time



...and also the mode age over time:

```
mod |>
  spread_draws(log_alpha[i], beta[i]) |>
  mutate(mode_age = 1/beta*log(beta/exp(log_alpha))+40) |>
  median_qi() |>
  mutate(year = years[i]) |>
  ggplot(aes(year, mode_age)) + geom_line() +
  geom_ribbon(aes(ymin = mode_age.lower, ymax = mode_age.upper), alpha = 0.2) +
  labs(title = "Estimated mode age at death over time", y = "age (years)")
```



5.2 Question

Can you forecast mortality rates with this model? If so, how? What are the assumptions behind the forecasts?

6 Partial data observed

Now we're going to switch gears a bit and have a look at the situation where we have mortality rates for five geographic areas, but in one area we only have partial information. This is based on simulated data (you can have a look to see how I generated it based on the `simulated_data.R` script).

For 4 areas, we have deaths and population counts from ages 40 up to 60:

```
df <- read_rds("/Users/liangqishen/Desktop/data/sim.rds")
df
```

```
# A tibble: 80 x 4
# Groups:   area [4]
   age area deaths pop
<int> <int> <int> <dbl>
```

```

1    40    1    1  5000
2    41    1    1  5000
3    42    1    1  5000
4    43    1    2  5000
5    44    1    1  5000
6    45    1    2  5000
7    46    1    0  5000
8    47    1    3  5000
9    48    1    5  5000
10   49    1    3  5000
# i 70 more rows

```

For one region, we just have ${}_{20}q_{40}$, that is, the probability of dying between ages 40 and 60:

```

q40 <- read_rds("/Users/liangqishen/Desktop/data/q40.rds")
q40

```

```
[1] 0.07262754
```

6.1 Relating ${}_{20}q_{40}$ to the Gompertz model

We want to fit a Gompertz model to each of the five areas (even the one with just a summary indicator). How to do this? Well, if we tell Stan how ${}_{20}q_{40}$ relates to μ_x , then the model has at least some information to estimate α and β . In particular, for each area, we're assuming:

$${}_1p_x = e^{-\mu_x}$$

where $\mu_x = \alpha e^{\beta x}$ and

$${}_{20}q_{40} = 1 - \prod_{x=40}^{60} {}_1p_x$$

Check the Stan file `gomp_partial.stan` to see this translated into code.

6.2 Fit the model

Get the data in the right format. Note that everything is a matrix now because we have more than one area:

```

y <- as.matrix(df |>
  pivot_wider(names_from = "area", values_from = "deaths") |>
  select(-age, -pop))
pop <- as.matrix(df |>
  select(-deaths) |>
  pivot_wider(names_from = "area", values_from = "pop") |>
  select(-age))

stan_data <- list(N = length(40:59),
  M = 4,
  K = 5,
  y = y,
  pop = pop,
  z = q40,
  age_c = (40:59)-40)

```

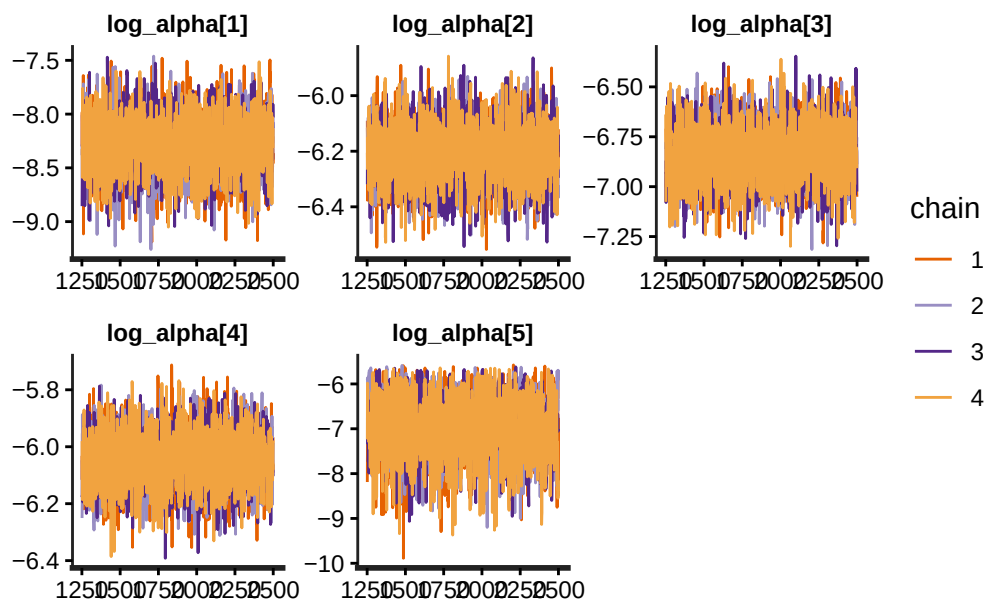
Fit the model:

```

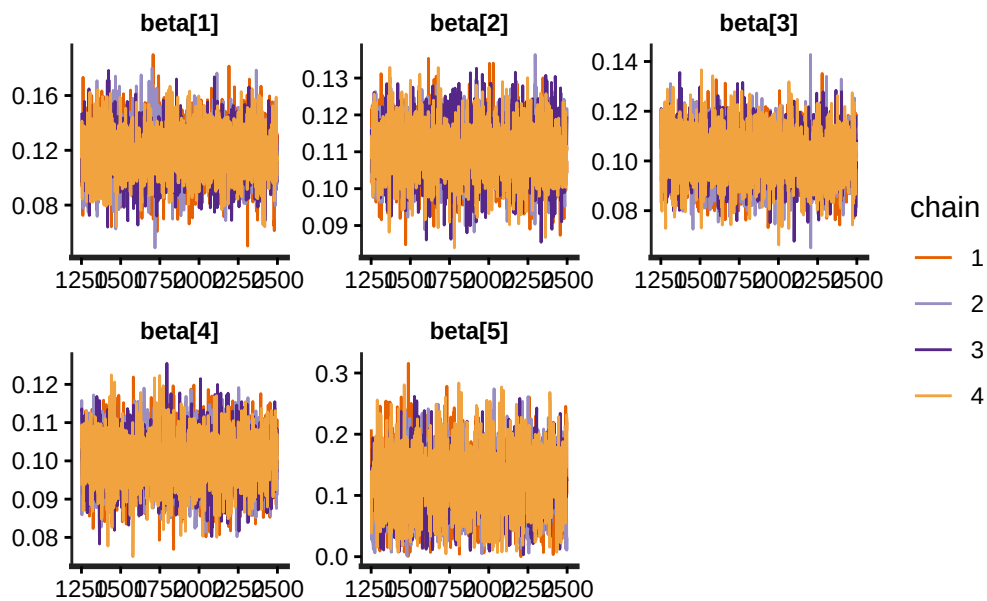
mod <- stan(file = "/Users/liangqishen/Desktop/data/gomp_partial_data.stan",
  data = stan_data,
  seed = 15,
  control = list(adapt_delta = 0.96),
  iter = 2500,
  refresh = 0)

traceplot(mod, c("log_alpha"))

```



```
traceplot(mod, c("beta"))
```

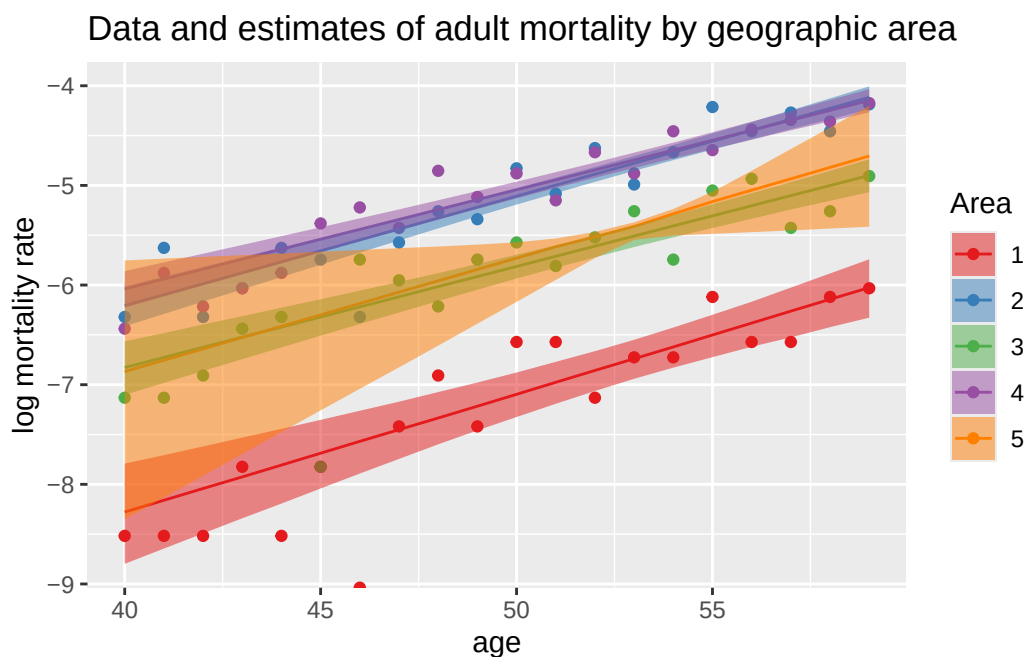


Let's plot the fitted lines (with uncertainty) with the observed data for each region. Notice the difference in uncertainty around area 5!


```

mod |>
  gather_draws(mu[i,j]) |>
  median_qi() |>
  mutate(age = i-1+40) |>
  rename(area = j) |>
  left_join(df) |>
  mutate(log_mx = log(deaths/pop)) |>
  ggplot(aes(age, log_mx))+
  geom_point(aes(color = factor(area)))+
  geom_line(aes(age, .value, color = factor(area)))+
  geom_ribbon(aes(age, ymin = .lower, ymax = .upper, fill = factor(area)), alpha = 0.5)+
  scale_color_brewer(name = "Area", palette = "Set1")+
  scale_fill_brewer(name = "Area", palette = "Set1")+
  labs(title = "Data and estimates of adult mortality by geographic area", y = "log mortality")

```



6.3 Compare parameter estimates against the truth

Here's what we estimated:

```

mod |>
  gather_draws(log_alpha[i], beta[i]) |>
  median_qi()

```

```
# A tibble: 10 x 8
```

	i	.variable	.value	.lower	.upper	.width	.point	.interval
	<int>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	1	beta	0.118	0.0830	0.155	0.95	median	qi
2	1	log_alpha	-8.28	-8.80	-7.79	0.95	median	qi
3	2	beta	0.110	0.0959	0.125	0.95	median	qi
4	2	log_alpha	-6.21	-6.41	-6.01	0.95	median	qi
5	3	beta	0.101	0.0819	0.121	0.95	median	qi
6	3	log_alpha	-6.83	-7.10	-6.57	0.95	median	qi
7	4	beta	0.0995	0.0862	0.113	0.95	median	qi
8	4	log_alpha	-6.04	-6.23	-5.86	0.95	median	qi
9	5	beta	0.114	0.0180	0.218	0.95	median	qi
10	5	log_alpha	-6.87	-8.35	-5.75	0.95	median	qi

And here's the truth (the values underlying the simulation):

```
read_rds("/Users/liangqishen/Desktop/data/true_params.rds")
```

```
# A tibble: 5 x 3
```

	are	log_alpha	beta
	<int>	<dbl>	<dbl>
1	1	-7.98	0.0985
2	2	-6.38	0.125
3	3	-6.88	0.110
4	4	-5.93	0.0924
5	5	-7.48	0.154